INGX2301 - LCIA (revised ch. 15 (Baksi, 2019))

Life cycle thinking is the prerequisite of any sound sustainability assessment. It does not make any sense at all to improve (environmentally, economically, or socially) one part of the system in one country or in one step of the life cycle if this "improvement" has negative consequences for other parts of the system which may outweigh the advantages achieved.

Walter Klopffer [1]

The sustainability assessment methods that we covered in previous chapters have included univariate footprint methods and thermodynamic aggregation methods. All these methods, except the carbon footprint method, focused on the use of resources such as water, fuel, and materials, while considering boundaries that range from including only human activities to including work done by nature as well. These methods do not consider emissions to the environment and their impact. The carbon footprint quantifies the potential impact of only greenhouse gases on climate change. In addition to resource consumption and greenhouse gas emissions, many other emissions such as volatile organic solvents, particulate matter, and ozone-depleting chemicals impact our well-being and that of the environment. The methods considered so far do not account for the impact of such emissions.
In his chapter we will learn about life cycle impact assessment (LCIA), which is the third step in the approach of life cycle assessment (LCA). We learned about the previous two steps of "goal definition and scope" and "inventory analysis" in Chapters 8 and 9. The overall approach of LCA is depicted in Fig 8.2. LCIA accounts for a large variety of emissions and resource use and combines many of the benefits of the sustainability assessment methods described in previous chapters. As we learned in Chapter 9 and saw in a typical inventory in Table 9.1, life cycle inventories for a selected product or process may consist of the emissions of hundreds of chemicals and use of a large number of resources. Determining the impact on the basis of such data is challenging not just because of the high dimensionality of the inventory, but also because of the need to associate emissions and resource use with human and ecological impacts. The footprint methods in Chapter 11 dealt with this "curse of dimensionality" by focusing only on selected flows such as greenhouse gases or water, while ignoring the rest. Methods based on energy and exergy analysis address the dimensionality challenge by representing resources in terms of the fuel value of fossil resources or the exergy of a more diverse array of resources. Cumulative exergy and energy analysis consider the work done in industrial and ecological systems for making various ecological goods and services available. However, none of these methods considers the large variety of molecules produced by human activities and their impact on human and ecological systems due to their release into society and the environment. Such impact assessment is an essential part of evaluating and developing sustainable solutions, but is a challenging task due to the thousands of chemicals in use today that may be emitted into the environment. Source: **Bakshi (2019)**

Life cycle impact assessment is a systematic approach for assessing the impact of emissions. It contains five steps, outlined in Fig. C1. The first three steps are mandatory in LCIA, in compliance with the ISO 14044 standard, while the second two, normalization and weighting, are optional. In this chapter we will learn about all five steps and their application. An overview of these steps for a typical LCIA approach is shown in Figure C2.
Source: **Hauschild and Huijbregts (2015)**

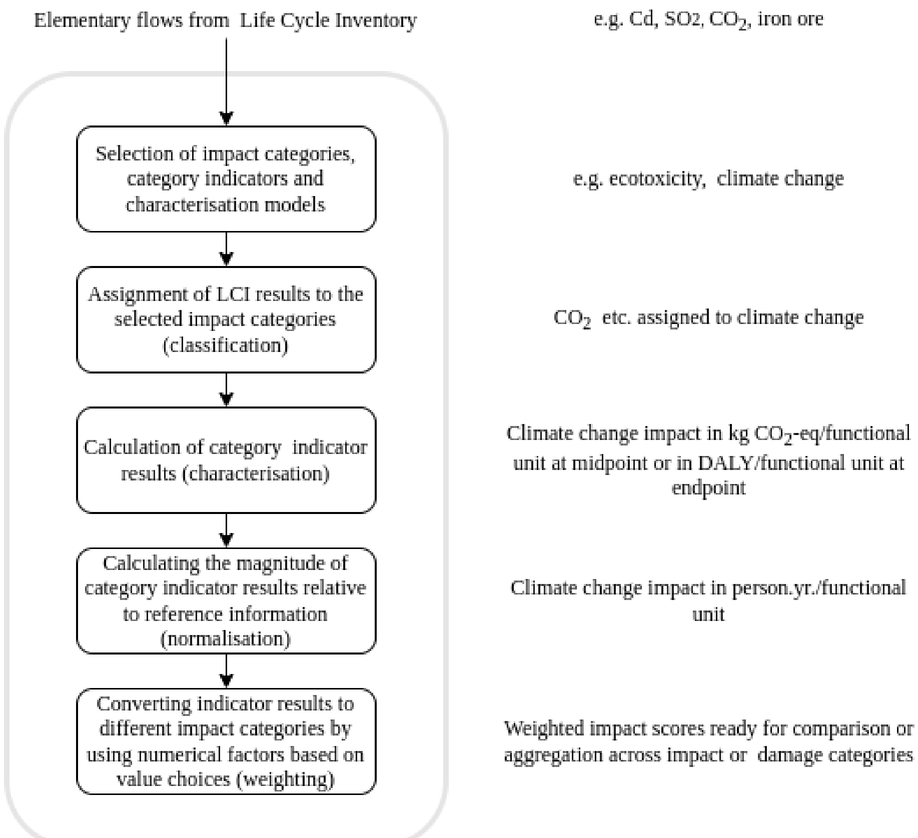

Figure C1: The five steps of life cycle impact assessment (**Hauschild and Huijbregts (2015)**).

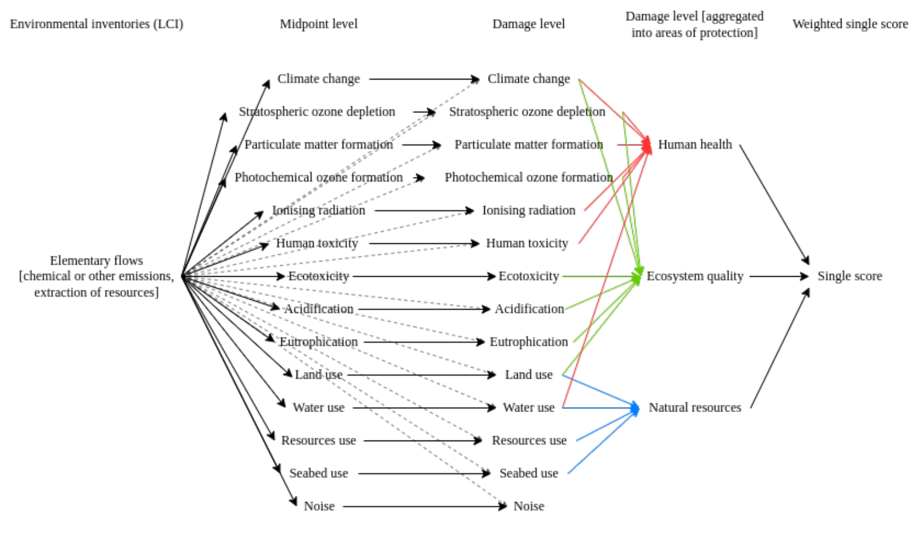

Figure C2: Updated LCIA framework (from **Verones et al. 2017**).

Steps in LCIA

Life cycle impact assessment consists of five chronological steps, each described as follows.

Step 1. Selection of impact categories, category indicators and characterization models (selection)

This is where the impacts to be addressed are selected in accordance with the goal of the study and a method chosen for each impact category. (Source: **Hauschild and Huijbregts (2015)**)

Step 2. Assignment of LCI results to the selected impact categories (classification)

The first step in reducing dimensionality in LCIA is to classify the emissions and resources into selected impact categories, such as those listed in Table 15.1 (see insert below).

Table 15.1 Typical classification categories in LCIA

Category	Cause and contributors	Unit
Climate change	Release of greenhouse gases such as CO_2, CH_4, NO_2, $CFCs$	kg CO_2
Ozone depletion potential	Emissions of gases that cause loss of stratospheric ozone such as chlorofluorocarbouns (HCFCs)	kg $CFC-11$
Photochemical smog formation	Reaction between NO_x and volatile organic compounds in the presense of sunlight to form ground-level ozone	kg CO_2-equivalent

Category	Cause and contributors	Unit
Human health polutants	Pollutants such as particulate matter and various toxics. These include hundreds of chemical compounds	kg $1,4$-dichlorobenzene equivalent
Ecotoxicity	Pollutants that are toxic to ecosystems including various chemicals	kg $1,4$-dichlorobenzene equivalent
Acidification	Addition of hydrogen ions (H^+) to the local environment. Caused by emissions such as NO_x, SO_x, NH_3	kg SO_2-equivalent
Eutrophication	Presense of excessive nutrients such as N and P compounds	kg N-equivalent
Land use	Area of land used	$m^2 \times y^{-1}$
Resource depletion	Fossil fuel	MJ
Resource depletion	Mineral use	kg antimony-equivalent

Here, inventory data, such as resource consumption and emissions into air or water are assigned to the relevant impact categories among those selected under step 1, according to their ability to contribute to different environmental problems. An example is given in Figure C3. Classification of various chemicals relies on knowledge about their environmental impact.

Such information has been developed through environmental and toxicological studies and in included in LCIA databases such as those listed in Table 9.3. Source: **Bakshi (2019)**

The left table in Figure C3 is example of a simplified inventory, showing what emission or resource we have, like CO2 emissions. In the second column, we see the quantity that is emitted per functional unit. The classification serves now for linking this inventory to the impact assessment categories. We allocate each emission to a category, like CO2 to climate change. Be aware of the fact, that one emission can have multiple relationships to impact categories. For example, NOx also has an influence on human toxicity and acidification, and so it is considered in both impact categories. None of the emissions is having an impact on ozone depletion.

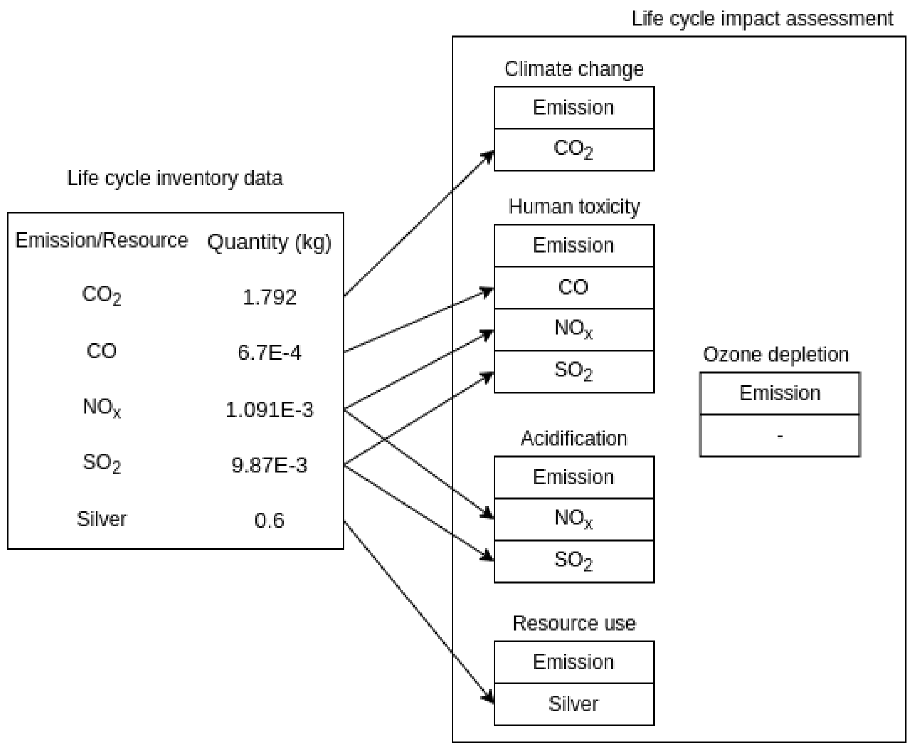

Figure C3: Classification example

Step 3. Calculation of category indicator results (characterization)

Inventory data classified in the same category are multiplied by their corresponding characterization factor to represent all flows of the common unit in the impact category. This step is represented by Equation 10.16, which is reproduced below:

$$h_i = \sum_j \phi_{ij} r_j$$

Here, h_i is the total impact for category i, r_j is the mass of inventory flow j, and ϕa is the characterization factor for impact category i and flow j.

This step represents the chemicals classified in the same group in terms of a common unit, to permit their addition. These common units are shown in Table 15.1 and typical characterization factors are shown in Table 15.2 and 15.7. The indicator scores for all the inventory data that contribute to a specific impact category are summed to arrive at a total impact score for that impact category.

There are two mainstream ways of deriving characterization factors: at midpoint or endpoint/damage level (Figure C2). Characterization factors at the midpoint level are located somewhere along the impact pathway, typically at the point after which the environmental mechanism is identical for all environmental flows assigned to that impact. Characterization factors at the damage/endpoint level correspond to three areas of protection, i.e. human health, ecosystem quality and resource scarcity. The two approaches are complementary in that the midpoint characterization has a stronger relation to the inventory data and a relatively low uncertainty, while the endpoint characterization provides better information on the environmental relevance of the inventory data, but is also more uncertain than the midpoint characterization factors. An example for the impact category climate change is given in Figure C4. Source: **Hauschild and Huijbregts (2015)**

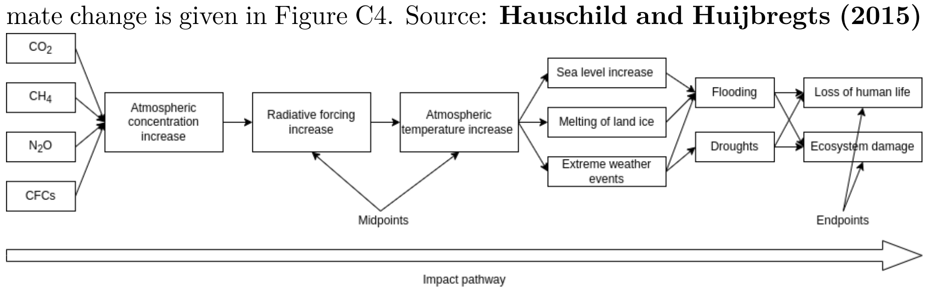

Figure C4: Simplified impact pathway for climate change connecting inventory data from the inventory to the areas of protection. (Source: Hauschild and Huijbregts (2015))

In order to facilitate the LCA practitioner's work with easy-to-apply LCIA methods, several authors have developed default lists of impact categories, often supported by default classification lists. A selection of several impact assessment methods is presented in Figure C5 and showing that the diversity and overlap in impact categories among different methods is huge. For simplification we will focus on the ReCiPe method.

ReCiPe was developed in 2008 by RIVM National Institute for Public Health and the Environment, CML, PRé Consultants and the Radboud University Nijmegen on behalf of the Dutch Ministry of Infrastructure and the Environment, with latest updates in 2016. It is a global impact method at midpoint and endpoint level. The full documentation is published on https://www.rivm.nl/en/life-cycle-assessment-lca/downloads.

Category	CML 2001	TRACI	Eco-indicator	ReCiPe	LC-impact	LIME	EDIP 2003	IMPACT 2002+	USEtox
Acidification	+	+	+	+	+	+	+	+	+
Climate change	+	+	+	+	+	+	+	+	+
Depletion of abiotic resources	+	+	+	+	+	+	-	+	+
Ecotoxicity	+	+	+	+	+	+	+	+	+
Eutrophication	+	+	+	+	+	+	+	+	+
Human toxicity	+	+	+	+	+	+	+	+	+
Ionising radiation	+-	-	+	+	+	+	-	+	+
Land use	+-	-	+	+	+	+	-	+	+
Odour	+-	-	-	-	-	-	-	-	-
Ozone layer depletion	+	+	+	+	+	+	+	+	-
Particulate matter	-	+	+	+	+	+	-	+	+
Photochemical ozone formation	+	+	-	+	+	+	+	+	-
Water use	-	-	-	+-	+	-	-	+	-

Figure C5: Overview of impact assessment methods (Adapted from: Acero et al. (2014)).

Step 3a. Characterization at midpoint level

A detailed overview of midpoint categories and related impact indicators in ReCiPe2016 is given in Table C1.

Table C1: Detailed overview of midpoint categories and related impact indicators in ReciPe2016.

Category	Indicator	Unit	CF_m	Abbr.	Unit
Climate change	Infra-red radiatinve forcing increase	$W \times y/m^2$	global warming potential	GWP	kg CO_2 to air
Ozone depletion	Stratospheric ozone decrease	$ppt \times y$	ozone depletion potential	ODP	kg CFC-11 to air
Ionizing radiation	Absorbed dose increase	$man - Sv$	ionizing radiation potential	IRP	kBq $Co - 60$ to air
Fine particular maatter formation	PM2.5 population intake increase	kg	particulate matter formation potential	PMFP	kg $PM2.5$ to air
Photochemical oxidant formation: ecosystem quality	Tropospheric ozone increase (AOTä40)	$ppb \times y$	photochemical oxidant formation potential - ecosystems	EOFP	kg NO_x to air
Photochemical oxidant formation: human health	Tropospheric ozone population intake increase (M5M)	kg	photochemical oxidant formation potential - humans	MOFP	kg NO_x to air
Terrestrial acidification	Proton increase in natural soils	$y \times m^2 mol/l$	terrestrial acidification potential	TAP	kg SO_2 to air
Freshwater eutrophication	Phosphorous increase in fresh water	$y \times m^3$	freshwater eutrophication potential	FEP	kg P to fresh water
Marine eutrophication	Dissolved inorganic nitrogen increase in marine water	$y \times kgO_2/kgN$	marine eutrophication potential	MEP	kg N to marine water
Human toxicity: cancer	Risk increase of cancer desease incidence	$-$	human toxicity potential	MTPc	kg $1,4DCB$ to urban air
Human toxicity: non-cancer	Risk increase of non-cancer desease incidence	$-$	human toxicity potential	MTPnc	kg $1,4DCB$ to urban air
Freshwater ecotoxicity	Hazard-weighted increase in fresh waters	$y \times m^3$	freshwater ecotoxicity potential	FETP	kg $1,4DCB$ to fresh waater
Marine ecotoxicity	Hazard-weighted increase in marine water	$y \times m^3$	marine ecotoxicity potential	METP	kg $1,4DCB$ to marine waater
Land use	Occupation and time-integrated transformation	$y \times m^2$	aggricultural land occupation potenial	LOP	$m^2 \times y$ annual crop land
Water use	Increase of water consumed	m^3	water consumption potential	WCP	m^3 water consumed
Mineral resource scarcity	Ore grade decrease	kg	surplus ore potential	SOP	kg Cu
Fossil resource scarcity	Upper heating value	MJ	fossil fuel potential	FFP	kg oil

Example 15.1 - following the textbook

Classify and characterize the following pollutants and resources from a selected life cycle inventory: 1.792 kg of CO_2, 6.7×10^{-4} kg of CO, 1.091×10^{-3} kg of NO_x, 9.87×10^{-4} kg of $S0_2$, and 0.6 kg of silver. The relevant characterization factors and mid-point indicators for these compounds are shown in Table 15.2.

Table 15.2 Illustration of the characterization step in LCIA

Emission/resource	Quantity (kg)	GWP	ODP	HT	AP	RD
CO_2	1.792	x1	-	-	-	-
CO	6.7E-4	-	-	x0.012	-	-
NO_X	1.091E-3	-	-	x0.078	x0.7	-
SO_2	9.87E-4	-	-	x1.2	x1	-

Emission/resource	Quantity (kg)	GWP	ODP	HT	AP	RD
Silver	0.6	-	-	-	-	x1.18
Total		1.792	0	2.04E-3	1.75E-3	7.1E-1

Note: GWP - global warming potential; *ODP* - ozone depletion potential; *HT* - human toxicity; *AP* - acidification potential; *RD:* resource depletion.

Solution The mid-point impact indicator for each flow may be calculated using Equation 10.16. For the human toxicity impact category, $h_i = (0 \times 1.792) + (0.012 \times 6.7 \times 10^{-4}) + (0.78 \times 1.091 \times 10^{-3}) + (1.2 \times 9.87 \times 10^{-4}) + (0 \times 0.6) = 2.04 \times 10^{-3}$ kg $1,4$-dichlorobenzene equivalent.

Or, through matrix notation

$$h = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0.012 & 0.78 & 1.2 & 0 \\ 0 & 0 & 0.7 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1.18 \end{bmatrix} \begin{bmatrix} 1.792 \\ 6.7 \times 10^{-4} \\ 1.091 \times 10^{-3} \\ 9.87 \times 10^{-4} \\ 0.6 \end{bmatrix} = \begin{bmatrix} 1.792 \\ 0 \\ 2.04 \times 10^{-3} \\ 1.75 \times 10^{-3} \\ 7.1 \times 10^{-1} \end{bmatrix} \begin{matrix} kg\ CO_2\ eq \\ kg\ CFC-11\ eq \\ kg\ 1,4DB\ eq \\ kg\ SO_2\ eq \\ kg\ Sb\ eq \end{matrix}$$

The solution is also shown in table 15.2.

```
# Computational implementation of what's above
import numpy as np

T = np.array([
    [1, 0, 0, 0, 0],
    [0, 0, 0, 0, 0],
    [0, 0.012, 0.78, 1.2, 0],
    [0, 0, 0.7, 1, 0],
    [0, 0, 0, 0, 1.18]
])
v = np.array([1.792, 0.00067, 0.001091, 0.000987, 0.6])

h = T@v
h = np.round(h,5)

print(*h, sep = '\n')
```

Example 15.2

Continuing Example 10.4 determine the mid-point indicators for producing 1 kWh electricity.

Solution Only two mid-point impact indicators are relevant to the emissions from these processes. The global warming potential of the life cycle is identical to the carbon footprint, and is 0.7 $kg\ CO_2\ eq./kWh$. Acidification potential is $(0.28 \times 1) + (0.13 \times 0.7) = 0.37\ kg\ S0_2\ eq.$

Choosing between alternatives based on mid-point impacts requires their comparison. Even with the smaller number of dimensions of mid-point indicators as compared to the original emissions data, this can still be a daunting task since the impacts are in diverse units. Furthermore, when comparing multiple products, only in rare cases do all the indicators for one product outperform the indicators for another product.

Example 15.3

Some mid-point indicators for various types of grocery bags are listed in Table 15.4 (see insert on next page). Choose the best option based on these results.

Solution In each column of Table 15.4, the smallest value in each category is indicated with italics. A close look shows that no bag is better in all categories. For example, on the basis of the global warming potential, the best bag is the reusable polypropylene bag, while on the basis of photochemical oxidant formation, the best bag is the biodegradable bag. From the numbers in the table, it is very difficult to identify the bag with the lowest impact. For easier visualization of the trade-offs. the indicators are plotted in Figure 15.2 after scaling the largest value to be 100. Here, the reference value for benchmarking is the largest value in each impact category.

Impact category	Unit	HDPE plastic bag 100% virgin	HDPE plastic bag with recycled content	Biodegradable bag	Oxo-degradable bag	Papear bag	Reusable PET bag	Reusable PP bag
GWP	$kg\ CO_2$eq	7.52	7.35	9.19	6.69	44.74	6.47	5.43
POCP	$kg\ C_2H_4$eq	0.045	0.038	-0.001	0.036	0.072	0.005	0.004
EP	$kg\ PO_4$eq	0.005	0.05	0.278	0.004	0.033	0.003	0.003
LU	Ha/yr	6.6E-6	5.9E-6	3.5E-4	9.2E-7	2.0E-3	2.0E-6	1.3E-6
WU	$kL\ H_2O$	0.013	0.053	0.050	0.012	0.423	0.038	0.016
SW	kg	2.74	3.31	0.83	2.56	3.24	0.81	3.27
Fossils	$MJ\ s$	19.93	19.07	9.96	17.94	44.77	6.46	6.65
Minerals	$MJ\ s$	8.44E-5	7.79E-4	1.02E-2	1.73E-2	7.42E-3	3.39E-5	2.50E-5

A glance at this plot shows that a paper bag has many more tall bars. Closer inspection reveals that it has the highest impact in all categories except eutrophication and mineral depletion. At the other extreme is the reusable PET bag owing to its lowest values in many categories. For categories in which the values for the reusable PET bags are not the smallest, the values are quite close to the lowest values. Thus, among all the bags considered, the reusable PET bag seems to be best. However, no conclusion can be reached unambiguously.

As this example illustrates, after applying various mid-point assessment methods, it is still common to face the challenge of having to choose between a higher impact in one category versus another. To address this challenge, end-point assessment methods have been developed for further aggregation to a smaller number of indicators, and even to a single indicator.

Step 3b. Characterization from midpoint to damage/ endpoint level

The framework and steps in end-point assessment are depicted in the right half of Figure C2. As shown, the mid-point impact categories are further aggregated into 3 areas of protection namely human health, ecosystem damage, and resource use by the following formula:

$$E_k = \sum_j \Theta_{j,k} h_j$$

Here $\Theta_{j,k}$ is the characterization factor that convert the j-th midpoint indicator h_j to an end-point indicator E_k Source: **Bakshi (2019)**.

Note that some Impact methods like LC Impact convert the life cycle inventory (LCI) directly to damage level (see for instance Figure C2; column 3, grey, dotted line) Example endpoint charaterization factors in the Recipe method are listed in Table 15.7.

Impact category	Unit	Individualist	Hierarchist	Egalitatian
Human health		-	-	-
Global warming	$DALY/kg\ CO_2$ eq.	8.12E-8	9.28E-7	1.25E-5
Stratospheric ozone depletion	$DALY/kg\ CFC$ eq.	2.37E-4	5.31E-4	1.34E-3
Ionizing radiation	$DALY/kBq\ Co-60$ emited to air eq.	6.80E-9	8.50E-9	1.40E-8
Fine particular matter formation	$DALY/kg\ PM2.5$ eq.	6.29E-4	6.29E-4	6.29E-4
Photochemical ozone formation	$DALY/kg\ NO_x$ eq.	9.10E-7	9.10E-7	9.10E-7
Toxicity (cancer)	$DALY/kg\ 1,4-DCB$ emited to urban air eq.	3.32E-6	3.32E-6	3.32E-6
Toxicity (non-cancer)	$DALY/kg\ 1,4-DCB$ emited to urban air eq.	6.65E-9	6.65E-9	6.65E-9
Water consumption	$DALY/m^3$ comsumed	3.10E-6	2.22E-6	2.22E-6
...		...	...	...

Before going into calculation we need to explain briefly the units of the charaterization factors (DALY, Species, and USD) and the social preferences (Individualistic, Hierarchist, Egalitarian).

HUMAN HEALTH

The indicator for the "Human Health" area of protection is "DALY", which stands for "Disablity adjusted life years". A DALY is the sum of years live disabled (YLD) and years of life lost (YLL) due to an impact (Figure C6). The concept is orginally developed by the World Health for use in health economics.

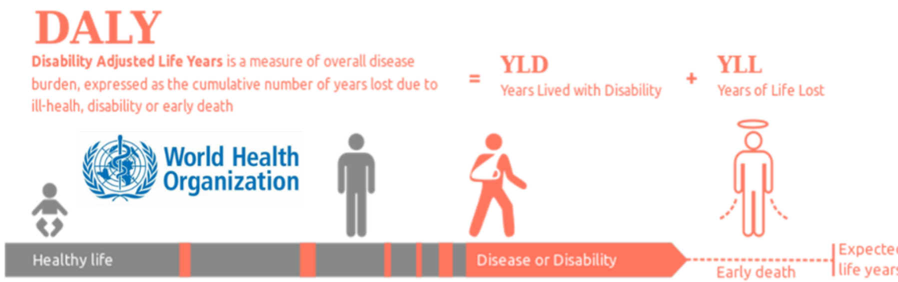

Figure C6: Concept of DALY (Source: Creative Commons License)

ECOSYSTEM DAMAGE

The endpoint unit recommended by the Life Cycle Initiative hosted by UN Environment for indicating damage to ecosytems is the "Potentially Disappeared Fraction of Species (PDF)". PDF is a relative measure of species loss and normally is calculated by diving the quantified species loss by the total number of present species in the corresponding region. Hence, the PDF lies between 0 and 1, where 1 means that all species are lost. However, as the PDF indicator is relatively new and previous methods only used the Species loss as indicator, we currently have characterization factors in both units. Therefore, ReCiPe decided to use species loss as endpoint indicator. To convert PDF to species loss, ReCiPe multiplies the PDF with the average species density for terrestrial or aquatic ecosystems

RESOURCE USE

Resource use is quantified in various ways. One approach, and this is the one used in Recipe, is to consider the increase in resource costs as function of depletion. The marginal costs increase (MCI) is the endpoint unit for resource extraction, calculated as the ratio of costs to the change in consumption.

SOCIAL PREFERENCES

In Recipe different sources of uncertainty and different choices were grouped into a limited number of perspectives or scenarios, according to the "Cultural Theory". These perspectives do not claim to represent archetypes of human behavior, they are merely used to group similar types of assumptions and choices. All three perspectives are included in ReCiPe and Table C2 provides an overview of how these perspectives were operationalized in the various impact categories.

The individualistic perspective is based on the short-term interest, impact types that are undisputed, and technological optimism with regard to human adaptation. An individualist considers nature to be capable of recovering from any disruption and is driven by short-term self-interest with high technological optimism. The hierarchist perspective is based on scientific consensus with regard to the time frame and plausibility of impact mechanisms. Hence, the hierarchists can be positioned in the middle. They consider nature to be capable of dealing with some disruption, and prefer that economic activity be kept within limits. The egalitarian perspective is the most precautionary perspective, taking into account the longest time frame and all impact pathways for which data is available. In other words, an egalitarian is almost the exact opposite of individualists and adopts the most precautionary approach with the longest time frame. It considers nature to be fragile and require us to use it carefully.

Source: **Bhavik R Bakshi** and **Hauschild and Huijbregts (2015)**

	Individualist	Hierarchist	Egalitarian
Climate change			
Time horizon	20 yers	100 years	1000 years
Climate-carbon feedbacks non-CO_2	No	Yes	No
Future socio-economic developments	Optimistic	Baseline	Pessimistic
Adaptation potential	Adaptve	Controlling	Comprehensive

Example 15.6

Represent the result from Example 15.1 in terms of endpoint indicators using the recipe method with hierarchist values. For silver, the surplus ore potential is 2000 $kg\ ore/kg\ Ag$, which may be converted into CU equivalents by the hierarchist value, 153 $kg\ Cu\ eq/kg\ ore$. Use the mid-point to end-point conversion factors in Table 15.7 to get the following results.

Solution By comparing the mid-point indicators in the last row of Table 15.2 with the impact categories in Table 15. 7 we can see that the impact relevant to human health are global warming and toxicity (non-cancer):

$$E_H = (1.792 \times 9.28 \times 10^{-7}) + (2.04 \times 10^{-3} \times 6.65 \times 10^{-9}) = 1.66 \times 10^{-6}\ DALY$$

The impact on terrestrial and freshwater ecosystems is due to global warming and acidification and may be calculated as follows:

$$E_E = ((2.80 \times 10^{-9} + 7.65 \times 10^{-14}) \times 1.792) + (2.12 \times 10^{-7} \times 1.75) = 5 \times 10^{-9} species \times years$$

The impact on resources is calculated separately for the use of mineral and fossil resources: $kgorekgCueq$

$$E_{Rm} = 0.6\ kg\ Ag \times 2000[kg\ ore/kg\ Ag] \times 153[kg\ Cu\ eq/kg\ ore] \times (2.31 \times 10^{-1}) = 4.24 \times 10^4\ USD2013$$

Step 4. Normalization

After the characterization step, the results of an LCA may include a large number of impact indicator scores, all expressed in individual metrics specific to the impact category. For example, climate change at midpoint level would typically be expressed in a unit of mass CO_2-equivalent. Similarly, photochemical ozone formation could be expressed in a unit of mass NMVOC-equivalent or mass C_2H_4-equivalent; see example 15.3. Also, at endpoint level, the different areas of protection are commonly expressed in different metrics and a comparison is not straightforward. Therefore, the interpretation of the results can be difficult if the LCA practitioner seeks to analyze the obtained characterized profile as a whole and identify the most relevant impacts to address in the decision-making process. To facilitate the interpretation of the LCA results and their communication to decision- and policymakers, two additional steps can be performed, namely normalization and weighting, which are both optional in the conduct of LCA. Normalization is addressed in this chapter; and weighting is discussed in Chapter 5. Normalization and weighting can either be performed on a midpoint or damage level. The preferred would be to do it on a damage level to get to a single indicator as proposed in Figure C2. Source: **Hauschild and Huijbregts (2015)**. However, as some LCA studies decide to stay on a midpoint level, normalization and weighing can also lead to "single midpoint indicator". A common normalization approach is to normalize each impact h_i by a reference value $h_{i,r}. Thus, the normalized value becomes:

$$\hat{h_i} = \frac{h_i}{h_{i,r}}$$

Common reference values include the total impact in a selected region, or the largest value among alternatives. Thus, the global warming potential from a product being analyzed would be divided by the global warming potential of emissions in the entire region where the product is produced, such as Recipe normalization factors according to this approach in Table C3.

	Midpoint				Endpoint			
	unit	World			unit	World		
Human health		Individualistic	Hierarchic	Egalitarian		Individualistic	Hierarchic	Egalitarian
Global Warming - Human health	kg CO2 eq. per person in 2010	1,08E+04	7,99E+03	5,80E+03	DALY per person in 2010	8,73E-03	7,42E-03	7,23E-02
Stratospheric ozone depletion - Human health	kg CFC11 eq. per person in 2010	6,53E-02	6,00E-02	7,84E-02	DALY per person in 2010	1,55E-05	3,19E-05	9,44E-05
Ionizing Radiation - Human health	kBq Co-60 emitted to air eq. per person in 2010	4,73E+02	4,80E+02	6,99E+02	DALY per person in 2010	3,13E-06	4,08E-06	9,78E-06
Fine particulate matter formation - Human health	kg PM2.5 eq. per person in 2010	1,60E+01	2,56E+01	2,56E+01	DALY per person in 2010	1,00E-02	1,61E-02	1,61E-02
Photochemical ozone formation - Human health	kg NOx eq. per person in 2010	2,06E+01	2,06E+01	2,06E+01	DALY per person in 2010	1,90E-05	1,90E-05	1,90E-05
Toxicity - Human health (cancer)	kg 1,4-DCB emitted to urban air eq. per person	9,90E-01	1,03E+01	2,95E+02	DALY per person in 2010	3,29E-06	3,42E-05	9,80E-04
Toxicity - Human health (non-cancer)	kg 1,4-DCB emitted to urban air eq. per person	5,09E+01	3,13E+04	2,22E+06	DALY per person in 2010	3,39E-07	2,08E-04	1,48E-02
Water consumption - human health	m3 consumed per person in 2010	2,67E+02	2,67E+02	2,67E+02	DALY per person in 2010	1,96E-04	1,96E-04	2,91E-04
Terrestrial ecosystems								
Global Warming - Terrestrial ecosystems	kg CO2 eq. per person in 2010	1,08E+04	7,99E+03	5,80E+03	Species.year per person in 2010	5,72E-06	2,31E-05	1,45E-04
Photochemical ozone formation - Terrestrial ecosy	kg NOx eq. per person in 2010	1,77E+01	1,77E+01	1,77E+01	Species.year per person in 2010	2,24E-06	2,24E-06	2,24E-06
Acidification - Terrestrial ecosystems	kg SO2 eq. per person in 2010	4,10E+01	4,10E+01	4,10E+01	Species.year per person in 2010	8,42E-06	8,42E-06	8,42E-06
Toxicity - Terrestrial ecosystems	kg 1,4-DBC emitted to industrial soil eq. per per	6,73E+03	1,52E+04	1,64E+04	Species.year per person in 2010	3,62E-04	8,19E-04	8,82E-04
Water consumption - terrestrial ecosystems	m3 consumed per person in 2010	2,67E+02	2,67E+02	2,67E+02	Species.year per person in 2010	0,00E+00	3,48E-06	3,48E-06
Land use - occupation	m2-annual crop eq per person in 2010	6,17E+03	6,17E+03	6,17E+03	Species.year per person in 2010	6,23E-06	6,23E-06	6,23E-06
Freshwater ecosystems								
Global Warming - Freshwater ecosystems	kg CO2 eq. per person in 2010	1,08E+04	7,99E+03	5,80E+03	Species.year per person in 2010	1,56E-10	6,11E-10	3,95E-09
Eutrophication - Freshwater ecosystems	kg P to freshwater eq. per person in 2010	6,50E-01	6,50E-01	6,50E-01	Species.year per person in 2010	4,90E-07	4,90E-07	4,90E-07
Toxicity - Freshwater ecosystems	kg 1,4-DBC emitted to freshwater eq. per perso	1,28E+01	2,52E+01	2,98E+02	Species.year per person in 2010	8,74E-09	1,75E-08	2,02E-07
Water consumption -aquatic ecosystems	m3 consumed per person in 2010	2,67E+02	2,67E+02	2,67E+02	Species.year per person in 2010	6,16E-10	6,16E-10	6,16E-10
Marine ecosystems								
Toxicity - Marine ecosystems	kg 1,4-DBC emitted to sea water eq. per person	8,80E+00	4,34E+01	2,48E+06	Species.year per person in 2010	9,24E-10	4,56E-09	2,59E-04
Eutrophication - marine ecosystems	kg N to marine water equivalents per person in	4,62E+00	4,62E+00	4,62E+00	Species.year per person in 2011	6,12E-09	6,12E-09	6,12E-09
Resources								
Mineral resource scarcity	kg Cu-eq per person in 2010	1,59E+05	1,20E+05	1,20E+05	USD2013 per person in 2010	3,08E+04	2,77E+04	2,77E+04
Fossil resource scarcity					USD2013 per person in 2010	2,93E+02	2,93E+02	2,93E+02
Crude oil	oil-eq per person in 2010	568,98	568,98	568,98				
Natural gas	oil-eq per person in 2010	0,48	0,48	0,48				
Hard coal	oil-eq per person in 2010	381,53	381,53	381,53				
Brown coal	oil-eq per person in 2010	31,46	31,46	31,46				
Peat	oil-eq per person in 2010							

Table C3: Normalization scores ReCiPe 2016

Another approach for normalization is to divide all numbers in an impact category by the largest quantity so that the largest value is equal to unity for all categories in order to estimate a benchmark for comparison across alternatives. The resulting dimensionless numbers allow convenient comparison. The normalized values of example 15.3 are shown in Figure 15.2, using values from Table 15.3. Both performance of normalization can fulfil two major purposes:

Table 15.3 Normalization in LCA

Impact category	Quantity, h_i	Normalization, $h_{i,r}$	Normalized value, $\hat{h}_i$
GWP	1.792	37.7E+12	47.5E-15
ODP	0	1E+9	0
HT	2.04E-3	576E+9	3.54E-15
AP	1.75E-3	286E+9	6.12E-15

The units of each category and its normalization factors are given in Table 15.1

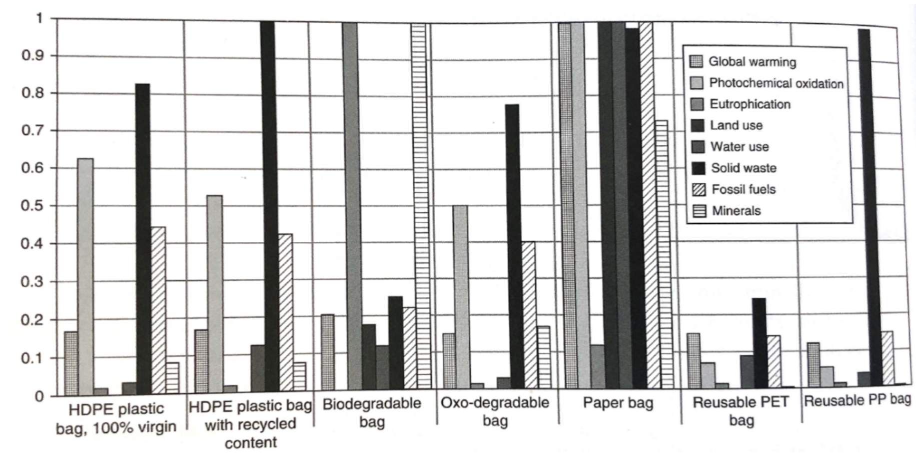

Figure. 15.2: Comparison of normalized indicators based on the data in Table 15.4.

1. To facilitate the interpretation and communication of the indicator results by (i) appraising the magnitude of the system's impact indicator scores relative to the impacts of the reference system, (ii) expressing the different impact indicator scores on a common scale, which supports comparisons of results across impact categories (keeping in mind that any differences in the severity or seriousness of the impact categories need to be dealt with separately in a weighting step), and (iii) preparing for weighting or valuation by expressing the scores in a form that is compatible with the intended weighting factors.

2. To check the sanity of the impact indicator scores, i.e. whether or not the results look reasonable, both in terms of the pattern across the categories ("is it reasonable that the normalized eutrophication impact is higher than the climate change impact?") and in the absolute value of the normalized results (e.g. the assessment of a pen leading to a climate footprint equivalent to the annual contribution of ten global average persons would reflect errors somewhere in the conduct of the LCA) Source: Hauschild and Huijbregts (2015)

Step 5. Weighting

The types of potential environmental impact associated with a product life cycle can vary widely from highly local, such as indoor air pollution and noise, to global, such as global warming and resource depletion, However, products or systems developed to improve the environment often focus on one or only a few environmental problems in order to alleviate or reduce their impact. This is the case in biofuels which are expected to help reduce the risk of global warming because their combustion is considered carbon neutral. However, while potentially reducing the risk of global warming, biofuels create a relatively greater impact to water resources by using crops as their raw materials and the occupation of land may impact biodiversity. Realizing that products and services are associated with diverse environmental impacts, we need to assess them by considering the balance among their environmental impacts in an explicit manner and in accordance with the defined goal of the study, to ensure that the conclusions take the whole relevant spectrum of impacts into account. Life cycle impact assessment covers multiple impact categories in the characterization phase. The ISO standard requires that "The selection of impact categories shall reflect a comprehensive set of environmental issues related to the product system being studied, taking the goal and scope into consideration. In Europe, a default list of 14 impact categories is recommended for studying environmental footprints, and where any one of these is excluded, validity of the exclusion needs to be explained. However, neither the midpoint nor the damage level support for the aggregation or comparison of scores for different impact categories. Therefore, if a trade-off between impact categories is created in a comparison of products, the final decision on which choice is preferable requires the use of some type of value judgment, i.e. a weighting process based on the perceived importance of the impact categories to the decision maker. How to weigh or balance the impact scores is left to the practitioner and the stakeholders involved in the study. It is important to recognize that the absence of assigning weights to impact scores results in equal weighting by default. Thus, the aim of the weighting step is to allow an aggregation of different environmental impacts into a single score. We can say it is a useful step because it resolves trade-offs in an explicit and transparent way in support of decision-making. Note: Weighting can only be performed after the normalization step (**Hauschild and Huijbregts (2015)**

Weighing factors for the midpoint are shown in Table 15.5 and for the endpoint in Table 15.8

Table 15.5: Typical midpoint weighting factors

Impact category	Weight
Resource depletion potential	0.361
Global warming potential	0.240
Photochemical oxidant chemical potential	0.113
Acidification potential	0.039
Human toxicity potential	0.039
Ecotoxicity, aquatic	0.106
Eutrophication potential	0.102

Table 15.8: Weights for combining impact categories by cultural theory

	Individualist	Egalitarian	Hierarchist	Average
Ecosystem quality	25	50	40	40
Human health	55	30	30	40

	Individualist	Egalitarian	Hierarchist	Average
Resources	20	20	30	20

Example 15.5

Choose between products A and B on the basis of the data in Table 15.6.

Table 15.6: Mid-point and end-point contributions of products A and B.

Category	Mid-point, A	Mid-point, B	Weight	End-point, A	End-point, B
GWP	0.14	0.25	0.240	3.4E-2	6.0E-2
POCP	0.55	0.72	0.113	6.2E-2	8.1E-2
AP	0.43	0.15	0.039	1.7E-2	5.9E-3
ET	0.4	0.3	0.106	4.2E-2	3.2E-2
Total				1.55E-1	1.79E-1

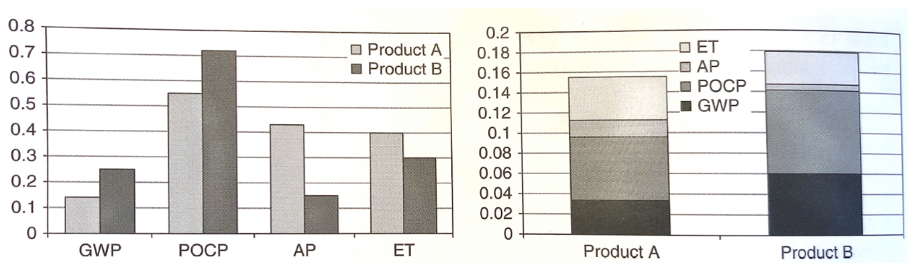

Figure 15.3: Left: normalized midpoint. Right: weighted single midpoint indicator

Solution

The mid-point indicators for products A and Bare plotted in Figure 15.3a. Choosing between the two products is difficult on the basis of these results since no product dominates in all categories. End-point indicators are shown in Figure 15.3b obtained using the weights in Table 15.5. Owing to the weighting scheme, the impact of photochemical oxidation dominates, followed by that of global warming. In terms of the end-point indicator, product A has a smaller impact and should be preferred. We have learned about methods for reducing the dimensionality from hundreds of flows in a typical life cycle inventory to about 20 mid-point indicators, to three AoP indicators, to a single indicator. The attraction of dimensionality reduction is quite obvious since it simplifies the task of choosing between alternatives and of making the results easier to communicate. However, there are some significant disadvantages of aggregation, which we learned about in Section 14.2.These are the implicit assumption of substitutability and the loss of information. When we convert multiple flows into a common unit, it is essential to use appropriate conversion factors for a fair comparison. The mid-point characterization factors represent such conversion factors. Even after proper conversion, aggregation implicitly assumes the flows to be substitutable. Also, by aggregation we lose details about individual flows, which can make it difficult to identify opportunities for improvement by changing these flows. There is also a strong trade-off between reducing dimensionality and increasing uncertainty in the results. As the dimensionality is reduced, the results become less certain. Source: **Bakshi (2019)**